

- Is fault-tolerant and provides high availability.
- Has the capability to share resources across multiple frameworks and implement a decentralised scheduling model.

Official website: <http://mesos.apache.org/>

Latest version: 1.3.0

Docker Engine

Docker Engine creates and runs Docker containers. A Docker container can be defined as a live instance of a Docker image, which is regarded as a file created to run a specific service or program in the operating system. Docker is an effective platform for developers and administrators to develop, share and run applications. It is also used to deploy code, test it and implement it as fast as possible.

Docker Engine is Docker's open source containerisation technology combined with a workflow for building and containerising applications. Docker containers can effectively run on desktops, servers, virtual machines, data centres-cum-cloud servers, etc.

Features

- Faster delivery of applications as it is very easy to build new containers, enabling rapid iteration of applications; hence, changes can be easily visualised.
- Easy to deploy and highly scalable, as Docker containers can run almost anywhere and run on so many platforms, enabling users to move the applications around — from testing servers to real-time implementation environments.
- Docker containers don't require a hypervisor and any number of hosts can be packed. This gives greater value for every server and reduces costs.
- Overall, Docker speeds up the work, as it requires only a few changes rather than huge updates.

Official website: <https://docs.docker.com/engine/>

Latest version: 17.06

Docker Swarm

Docker Swarm is an open source native clustering tool for Docker. It converts a pool of Docker hosts into a single, virtual Docker host. As Docker Swarm serves the standard Docker API, any tool communicating with the Docker daemon can use Swarm to scale to multiple hosts.

The following tools support Docker Swarm:

- Dokku
- Docker Compose
- Docker Machine
- Jenkins

Unlike other Docker based projects, the 'swap, plug and play' principle is utilised by Swarm. This means that users can swap the scheduled backend Docker Swarm out-of-the-box with any other one that they prefer.

Features

- Integrated cluster management via Docker Engine.
- *Decentralised design:* Users can deploy both kinds of nodes, managers and workers, via Docker Engine.
- Has the Declarative Service Model to define the desired states of various services in the application stack.
- *Scaling:* Swarm Manager adapts automatically by adding or removing tasks to maintain the desired state of application.
- Multi-host networking, service discovery and load balancing.
- Docker Swarm uses TLS mutual authentication and encryption to secure communications between the nodes.
- It supports rolling updates.

Official website: <https://docs.docker.com/engine/swarm/>

Version: Supports Docker Engine v1.12.0 (the latest edition)

Kontena

Kontena is an open source project for organising and running containerised workloads on the cluster and is composed of a number of nodes (virtual machines) and Master (for monitoring nodes).

Applications can be developed via Kontena Service, which includes the container image, networking, scaling and other attributes of the application. Service is highly dynamic and can be used to create any sort of architecture. Every service is assigned a DNS address, which can be used for inter-service communication.

Features

- In-built private Docker image registry.
- Load balancing service to maintain operational load of nodes and Master.
- Access control and roles for secure communication.
- Intelligent scheduler with affinity filtering.

Official website: <https://www.kontena.io/>

Latest version: 1.3.3

oVirt

oVirt is an open source virtualisation platform, developed by Red Hat primarily for the centralised management of virtual machines, as well as compute, storage and networking resources, via an easy Web based GUI with platform independent access. oVirt is built on the powerful Kernel-based Virtual Machine (KVM) hypervisor, and is written in Java and the GWT Web toolkit.

Components

- *oVirt Node:* Has a highly scalable, image based and small footprint hypervisor written in Python.
- *oVirt Engine:* Has a centralised virtualisation management engine with a professional GUI interface to